Bocconi

CORPORATE FINANCE 30017

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Applying the CAPM (Advanced)

Università Bocconi
MILANO

Agenda

RECAP

WRDS BETA SUITE

UNCERTAINTY IN BETA ESTIMATION

RECAP

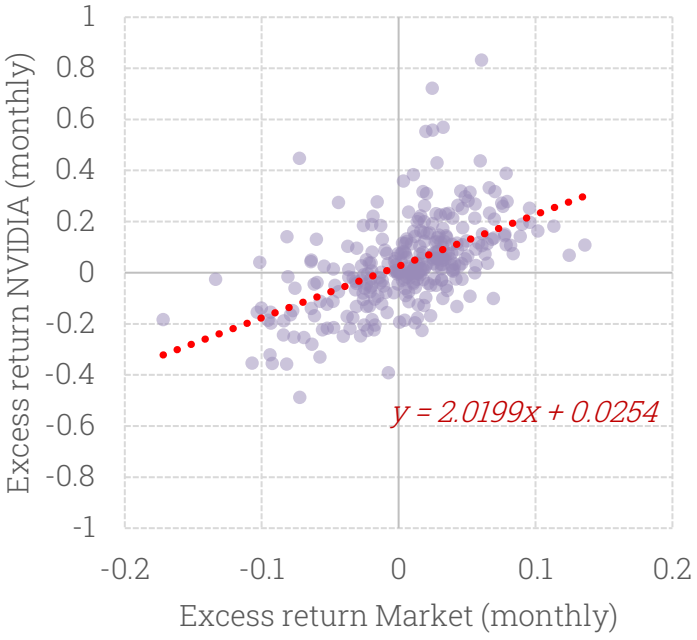
Estimate β_i

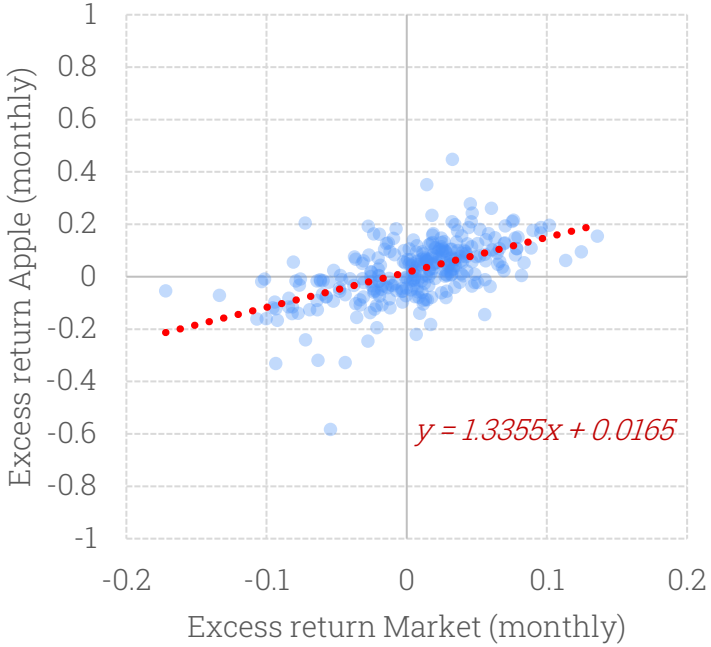
- Define excess returns (to ease notation):
  - $y_t = r_{i,t} - r_{f,t}$
  - $x_t = r_{m,t} - r_{f,t}$
- Univariate regression:

$$y_t = \alpha_i + \beta_i x_t + e_t$$

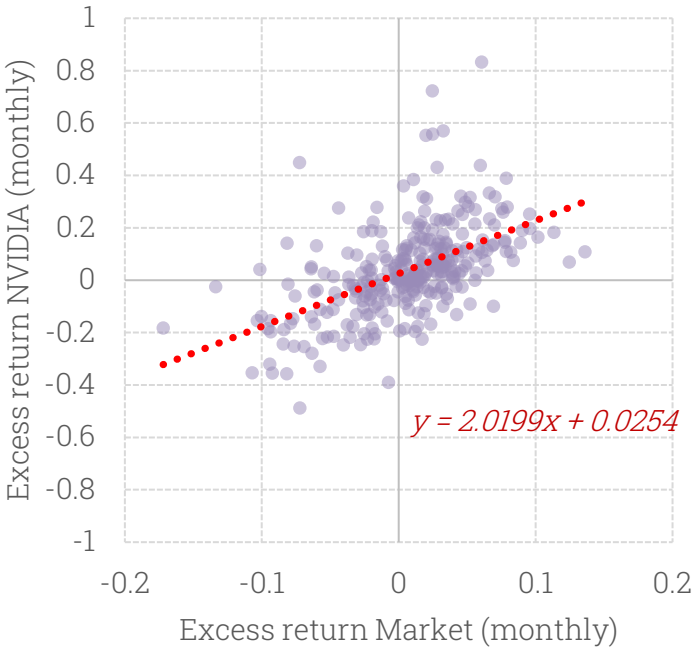
$$\widehat{\beta_i} \xrightarrow{\mathrm{T}\to\infty} \beta_i = \frac{Cov\left(r_{i,t} - r_{f,t}, r_{m,t} - r_{f,t}\right)}{Var\left(r_{m,t} - r_{f,t}\right)}$$

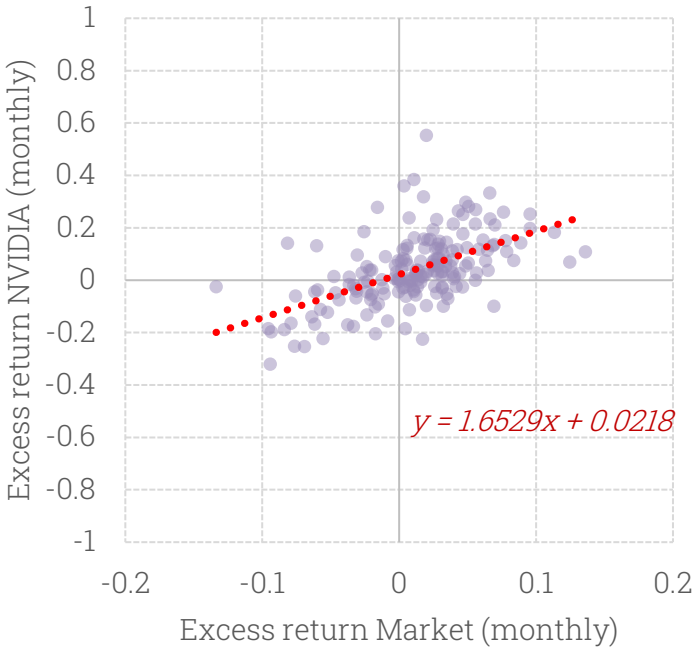
Apple vs. NVIDIA





NVIDIA 2000-2024 vs. 2010-2024

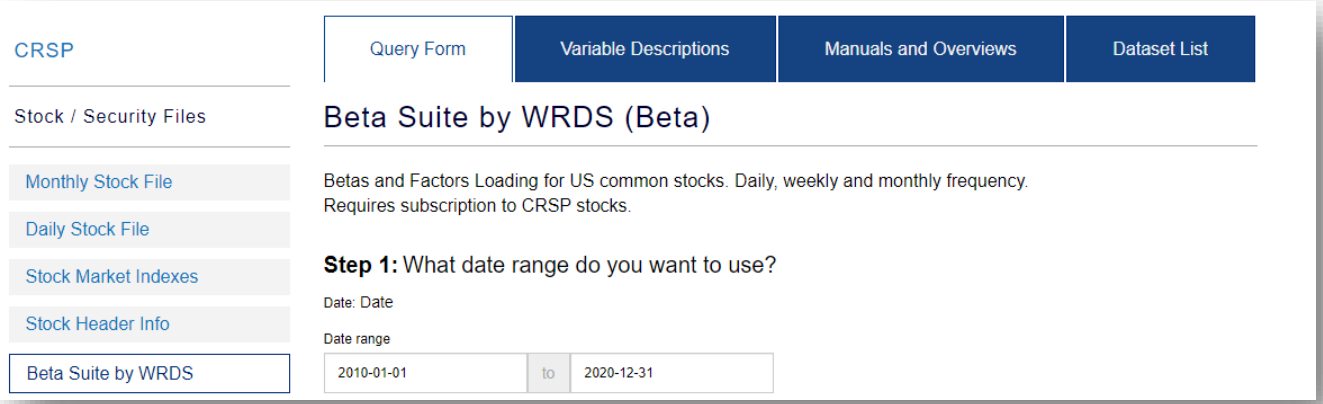




WRDS BETA SUITE

WRDS Tool to Estimate β_i – *fast & scalable*

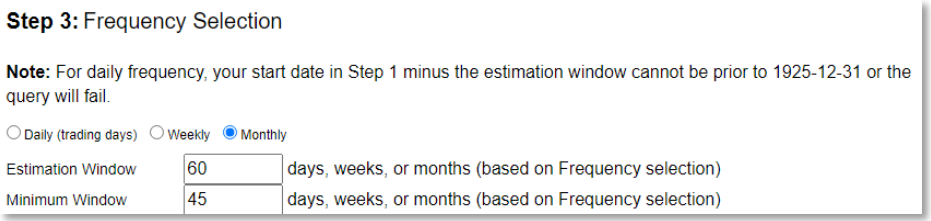
- Estimates single factor (i.e. market model) and multi-factor (i.e. Fama French 3 factor, Fama French 4 factor, Scholes-Williams) betas
- Produces rolling beta estimates, that for a given time t use the corresponding past time series
- Daily, weekly, monthly estimates



Apply WRDS Beta Suite

Log into WRDS, navigate to *Beta Suite by WRDS*

1. Select period 2000 to 2020
2. Select NVIDIA, Apple
   - *Alternative: Upload PERMNOs of SP500 index constituents as of 31dec2020, for example (scale!)*
3. Select Monthly returns, 5 year estimation window
4. Market model, regular returns
5. Select Ticker, Permno



WRDS runs thousands of regressions for us, since these are monthly, firm-level rolling…

Thousands of β_is..

For each company:

- Inspect time series of β_i
- Inspect distribution of β_i

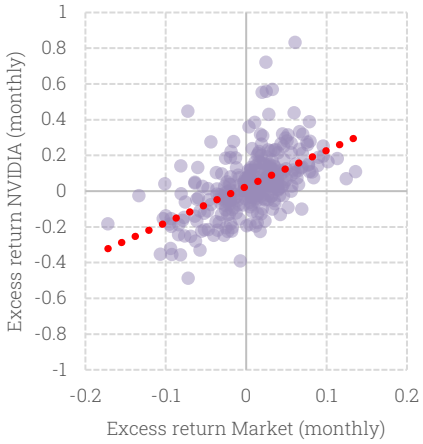
- Inspect time series of a_i
- Inspect distribution of a_i

Can also inspect other regression diagnostics…

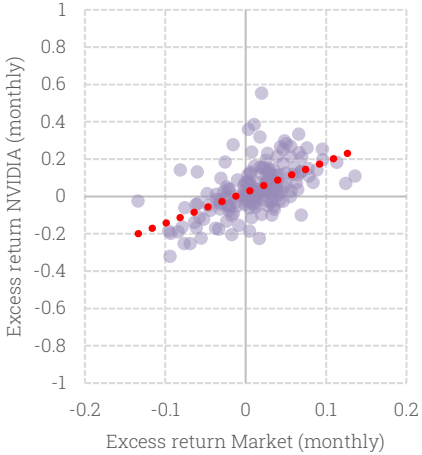
	PERMNO	DATE	n	RET	b_mkt	alpha	ivol	tvol	R2	exret	TICKER
1	10104	29jan2010	60	-.05788835	.935139	.009149	.05079861	.06728999	.43009462	-.02690187	ORCL
2	10104	26feb2010	60	.06895057	.955105	.011086	.04957751	.0670964	.45402737	.03715583	ORCL
3	10104	31mar2010	60	.04300201	.937407	.011143	.04954786	.06690501	.45155571	-.01736514	ORCL
4	10104	30apr2010	60	.00807084	.921529	.01185	.04894788	.06590283	.44835513	-.0107773	ORCL
5	10104	28may2010	60	-.12747659	.943596	.009748	.04905025	.06726684	.46828316	-.05486794	ORCL
6	10104	30jun2010	60	-.04918035	.945417	.009412	.04902319	.0676457	.47480225	.00327795	ORCL
7	10104	30jul2010	60	.10391428	.9648	.010251	.04907565	.06872651	.49010173	.0382969	ORCL
8	10104	31aug2010	60	-.07593063	.974109	.010327	.04901556	.06925035	.4990159	-.03000969	ORCL
9	10104	30sep2010	60	.22911426	1.05155	.013448	.05075668	.07440592	.53465888	.13608422	ORCL
10	10104	29oct2010	60	.09608934	1.06513	.013664	.05091403	.07515498	.54105637	.05518919	ORCL
11	10104	30nov2010	60	-.07947581	1.07564	.013019	.05191933	.07608372	.53433379	-.08596661	ORCL
12	10104	31dec2010	60	.15733035	1.10328	.014884	.0523575	.0780189	.54964173	.08387162	ORCL
13	10104	31jan2011	60	.02492011	1.1059	.015053	.05229887	.07801682	.55062572	.0028648	ORCL
14	10104	28feb2011	60	.02716212	1.09969	.015077	.05229367	.07790958	.54947743	-.01153378	ORCL
15	10104	31mar2011	60	.01618533	1.09456	.013898	.05153973	.07718582	.55412894	.01102675	ORCL
16	10104	29apr2011	60	.07739476	1.09854	.013745	.05143838	.07736367	.55792046	.04565257	ORCL
17	10104	31may2011	60	-.04838704	1.10224	.013001	.05181092	.07759146	.55412335	-.03443557	ORCL
18	10104	30jun2011	60	-.03828175	1.10648	.012358	.05196813	.07791224	.55509955	-.01899258	ORCL
19	10104	29jul2011	60	-.06897599	1.11697	.010985	.05234022	.07864804	.55711062	-.0429737	ORCL
20	10104	31aug2011	60	-.0820798	1.12568	.010399	.05243554	.07952572	.56525333	-.01527345	ORCL
21	10104	30sep2011	60	.0238689	1.07158	.010404	.0523954	.07805192	.54937096	.10930774	ORCL
22	10104	31oct2011	60	.1423104	1.07923	.0107	.0523996	.079831	.56916311	.02068622	ORCL
23	10104	30nov2011	60	-.04333232	1.08038	.009882	.05281168	.08011971	.56550866	-.04031049	ORCL
24	10104	30dec2011	60	-.1818182	1.07847	.008614	.05660943	.08259269	.53022025	-.18981301	ORCL
25	10104	31jan2012	60	.10214423	1.09066	.009698	.05671412	.08343626	.53796735	.04768132	ORCL
26	10104	29feb2012	60	.0370436	1.08193	.009949	.05659689	.08315684	.53677794	-.01116359	ORCL
27	10104	30mar2012	60	-.00324728	1.07311	.007828	.05587386	.08239562	.54015742	-.03689517	ORCL
28	10104	30apr2012	60	.01028806	1.07469	.00823	.05586808	.08234614	.53969989	.0194095	ORCL
29	10104	31may2012	60	-.09965988	1.09105	.007803	.0560789	.08353247	.54929914	-.03323639	ORCL
30	10104	29jun2012	60	.12202499	1.10805	.008531	.05673322	.08481181	.55253171	.07958322	ORCL
31	10104	31jul2012	60	.01885522	1.10787	.008577	.05673338	.08462039	.55050249	.01010159	ORCL
32	10104	31aug2012	60	.04801321	1.10788	.008133	.05653732	.08454969	.55285693	.01966264	ORCL
33	10104	28sep2012	60	-.00600318	1.09881	.007071	.05675728	.0842833	.54651796	-.03634817	ORCL
34	10104	31oct2012	60	-.01017162	1.09912	.007204	.05675228	.08430941	.54687853	.00906743	ORCL
35	10104	30nov2012	60	.03523164	1.08676	.008349	.05645837	.08329033	.54051906	.0265585	ORCL
36	10104	31dec2012	60	.04118106	1.09507	.006699	.05438762	.0822446	.56269399	.02825729	ORCL
37	10104	31jan2013	60	.06572625	1.08377	.007208	.05425092	.0814434	.55628677	.00473085	ORCL
38	10104	28feb2013	60	-.03576448	1.06957	.007344	.05418606	.0806564	.54866663	-.04974508	ORCL
39	10104	28mar2013	60	-.0557827	1.0538	.004989	.05560141	.08107812	.52971154	-.09888651	ORCL
40	10104	30apr2013	60	.01391887	1.0511	.004704	.05559133	.08078905	.52651189	-.00241499	ORCL
41	10104	31may2013	60	.03050641	1.04602	.003519	.05483999	.08010446	.53131463	.00107571	ORCL

UNCERTAINTY IN BETA ESTIMATION

Structural Change vs. Statistical Noise



y = 2.0199x + 0.0254



y = 1.6529x + 0.0218

Beta estimates change across time window:

1. True relationship with market may evolve (*why?*)
2. Random shocks may distort estimate (*why?*)

$$y_t = \alpha_i + \beta_{i,t} x_t + e_t$$

How to tell both drivers apart?

Role of Standard Errors

$$\sqrt{T}(\widehat{\beta}_i - \beta_i) \xrightarrow{d} N\left(0, \frac{Var(e_t)}{Var(x_t)}\right)$$

- Standard errors $SE(\widehat{\beta}_i)$ are the square root of the estimate of the estimator's (limit) variance
- They quantify scope of statistical noise (due to e_t)
- Allow construction of confidence intervals:

$$[\widehat{\beta}_i - z_q \times SE(\widehat{\beta}_i);\ \widehat{\beta}_i + z_q \times SE(\widehat{\beta}_i)]$$

- Helps to assess whether estimates from two periods are "statistically different"
  - If yes, then structural change is likely driver of difference

Standard Deviation vs. Standard Error

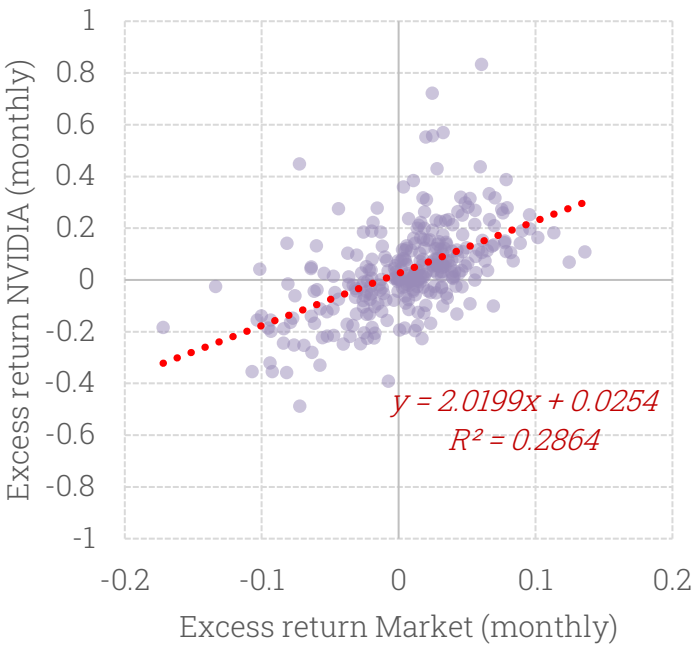
- Returns (and other data) vary:
  - Captured by **Standard Deviation**
- Variation unexplained by statistical model renders estimates uncertain:
  - Captured by **Standard Error**
  - *In excel, use =LINEST() to get regression standard errors*

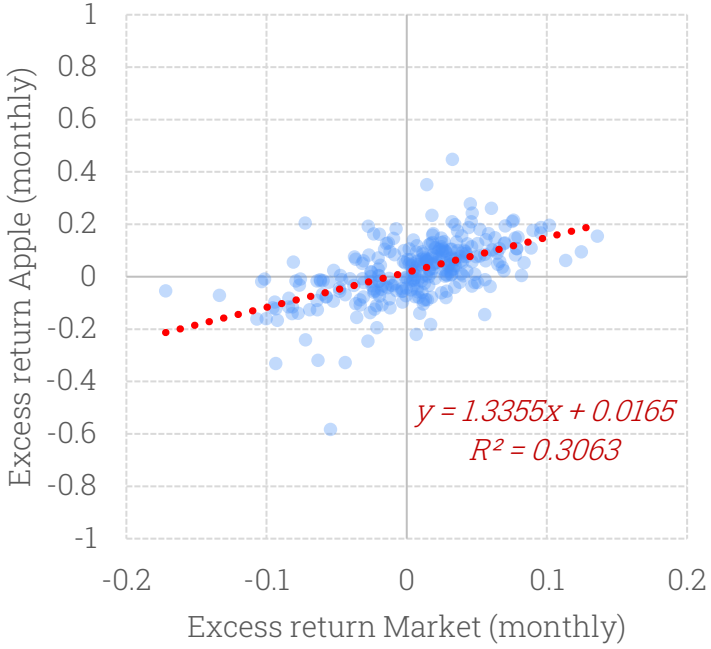
CAPM Regression Error = Idiosyncratic Shocks

$$y_t = \alpha_i + \beta_i x_t + \boldsymbol{e_t}$$

- Recall:
  - $y_t = r_{i,t} - r_{f,t}$
  - $x_t = r_{m,t} - r_{f,t}$
- α_i captures expected excess return unrelated to risk factors
- $\beta_i x_t$ captures variation in excess return due to variation in market excess returns
- $\boldsymbol{e_t}$ captures how realized excess returns deviate from market evolution and α_i
  - Such deviations are driven by idiosyncratic (= asset specific) risk factors (through lens of CAPM)

Where do you see more idiosyncratic risk?





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Regression diagnostic R^2 captures this formally.

Understanding Risk

	Standard Deviation (%)	R^2	Beta	Standard Error of Beta
Sun Life Financial	18.7	0.12	0.86	0.30
Loblaw	19.5	0.06	0.63	0.33

A) Which stock is *riskier*?

B) What proportion of SLF's risk was *market risk,* what was *idiosyncratic risk*?

C) Suppose market provides 5% excess return next year, alpha is zero. What can you expect for excess returns of:

- Sun Life Financial?
- Loblaw?

D) What is the confidence interval for your predictions?

Obtain confidence intervals

1. Obtain confidence interval for β_i from data exercise of last session
   - In Excel use the =LINEST() function to obtain standard errors
     - Verify that the slope coefficient is the same as in the scatterplot. Also compare to =SLOPE()
     - Feel free to use any statistical software package (Stata, R, Python Pandas, Eviews, Matlab, SAS,…)
   - Compute the 95% confidence interval $[\beta_l;\ \beta_h]$ using

$$\beta_l = \hat{\beta} - 1.96 \times SE(\hat{\beta})$$
$$\beta_h = \hat{\beta} + 1.96 \times SE(\hat{\beta})$$

2. Compute 95% confidence interval for excess return prediction (assuming zero α_i) as

$$\left[\beta_l \times \boldsymbol{E}[r_{m,t} - r_{f,t}];\ \beta_h \times \boldsymbol{E}[r_{m,t} - r_{f,t}]\right]$$